

University of Oxford, Hilary term 2026, Syllabus for: Foundations of machine learning

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class time	Thursday 15:00-16:30 and Friday 11:30-13:00
webpage	https://maxkasy.github.io/home/ML-Oxford_2025/
canvas page	https://canvas.ox.ac.uk/courses/294344
location	Lecture Theatre, Manor Road Building

1 Overview and Objectives

The goal of this course is to provide you with solid **theoretical foundations in machine learning**. This will allow you to

1. become critical consumers of machine learning research, including an understanding when new methods might or might not be useful for empirical work in economics,
2. develop your own research agenda around importing ideas from machine learning into economic and econometric theory, and
3. speak to the machine learning literature, contributing ideas from economics.

We begin by introducing some **foundations**. The class starts with a review of statistical decision theory, which provides the conceptual framework for the rest of the course. We then introduce basic ideas of supervised learning, using *Python* packages for machine learning. We next introduce probably approximately correct learning theory for classification and prediction. We then consider regularization and data-driven choice of tuning parameters. We will discuss the canonical normal means model. In this model, we will motivate shrinkage estimators in different ways, and will prove the famous result that shrinkage estimators can uniformly dominate conventional estimators.

We will then apply these general ideas and discuss several methods for **supervised machine learning**, that is, prediction. We will discuss Gaussian process regressions, random forests, deep neural nets, and transformer models for natural language processing. In this context, we will also consider numerical methods used for training neural nets, such as stochastic gradient descent. We finish this part of class by discussing conformal inference, a simple general technique to calculate valid confidence intervals for the predictions of any supervised learning algorithm.

The next part of class will cover different frameworks for **online and adaptive learning**. We will start with the adversarial online learning setting, where no probabilistic assumptions about data generation are made at all, and we will explore the connection of statistical learning to (online, convex) optimization. We will next consider multi-armed bandits, and review some theoretical results providing performance guarantees (regret bounds) for algorithms used for learning in bandit settings. We will then turn to a generalization of bandit problems, Markov decision problems, and will discuss reinforcement learning approaches for solving these.

The class will conclude with a discussion of **ethics** and the **social impact** of artificial intelligence. We will, in particular, review debates surrounding fairness and discrimination, as well as differential privacy, and explainability.

If you need any special accommodations for the lectures, for physical or medical reasons, please send me an email.

2 Assignments

Your grade for this class will be based on four assignments. (The overall workload will be the same as in previous years, but with more in-person components.)

1. A portfolio of 4 coding exercises, **due on 7 April 2026** (30% of grade).
2. A research proposal of 1500-2000 words, **due on 7 April 2026** (30% of grade).
3. A 5 minute in-class presentation of your research proposal at the end of term, to take place in **Week 8** of HT (10% of grade).
4. A 90 minute exam after the end of term, which covers the theoretical background and interpretation for the coding exercises, to take place in **week 9** of HT (30% of grade).

Both the coding portfolio and the research proposal have to be submitted in **pdf** format via **Inspira**.

Coding portfolio The portfolio of coding exercises will be based on four problem sets. The problem sets focus on implementing some of the methods discussed in class in **R**, or in **Python** (the choice is yours, except for the first exercise, which uses Python). You will conduct simulation exercises to verify some of our theoretical results numerically. To help you learn these programming languages, if needed, I have compiled a list of resources here: <https://maxkasy.github.io/home/computationlinks/>.

I strongly encourage you to complete the problemsets early during the term, to avoid a time-crunch at the end of term, and to stay in sync with lectures. The suggested completion dates for the problemsets are 31 January, 14 February, 28 February, and 14 March. Your coding portfolio has to satisfy the following conditions:

- For each of the 4 assignments, all results should be generated from a single script, running from start to end, producing all the output.

- Output and discussion of findings have to be integrated in a report generated in R-Markdown or Quarto (for R), or a Jupyter notebook (for Python).
- Figures and tables have to be clearly labeled and interpretable.
- You have to submit the pdf (and only the pdf) generated from your script. This pdf should contain, first, a brief discussion and overview of your code. The pdf should then show the source code used to generate your findings. This pdf should, finally, show the figures and tables generated by your code, joint with a brief discussion of your findings.
- Please make an effort to make this “human-readable” and do not include lists of warnings, intermediate calculations, etc.

You are allowed, and in fact encouraged, to collaborate in small groups (of up to 4 students) for the coding assignments. You are allowed to share jointly written computer code with the classmates in your group. However, each one of you must write up your answers completely independently.

Research proposal The research proposal has to be based on one of the areas of machine learning covered in class. Proposals might fall in one of several domains, including

- (i) applications of machine learning in empirical economics,
- (ii) econometric theory drawing on ideas from machine learning, and
- (iii) machine learning theory, drawing on ideas from economics.

Your research proposals have to satisfy the following conditions:

1. They cannot be identical with the topic of the MPhil thesis that you might be writing concurrently.
2. The proposals have to indicate clearly whether your contribution is of type (i), (ii), or (iii), as described above.
3. The proposals should be in 12pt font, 1.5 linespacing, margins of 3cm, A4. The length should be 1500-2000 words. Code and references do not count to this word limit.
4. The proposals cannot consist exclusively of a summary of literature; they have to propose new research.

Presentation In the last week of class, you will give a presentation of your research proposals.

- The proposal does not need to be finished at this point. The presentation is an opportunity to obtain feedback from me and your classmates.
- The presentation should be no longer than 5 minutes.
- I recommend no more than 3 slides, at most 2 formulas per slide, and no more than 10 numbers per slide (if you have simulations or empirical results).

Exam The 90 minute exam will focus on the theoretical background for the four coding portfolios.

- This substitutes for theoretical discussions that were previously part of the coding exercise.
- The exam will be computer-based and will take place in the exam schools.
- More detail will be provide in class towards the end of term.

Policy on the use of generative AI The department policy on generative AI applies:

When undertaking study and research for an MPhil course or thesis you are permitted to make selective use of AI in your work for tasks such as the following:

- *Brainstorming ideas*
- *Surveying literature relevant to your area of study*
- *Gathering data and generating code to implement estimation techniques in econometric work*
- *Generating diagrams, charts and results tables for inclusion in submissions and/or the thesis*
- *Proof-reading and obtaining feedback on text that you have written*

It is important that you use AI as an enabler of your work rather than relying on AI to generate the intellectual value added in your submitted work/ thesis; in particular, the decisions that you take on how to implement research methods and the conclusions that you draw from them should be based on your own reasoning.

In order to promote fair use of AI the University provides all students with access to tools developed by OpenAI. Before making use of these tools please read the University guidelines on the use of AI available at the following link: <https://www.ox.ac.uk/students/academic/guidance/skills/ai-study>

When students submit an assignment or thesis on Inspira they are asked to complete a Declaration of Authorship. For Economics submissions this will include a question that asks candidates to provide a clear statement of the different tasks that they have completed with the assistance of AI. The information provided by candidates will be shared with examiners and thesis assessors.

3 References

Required readings are marked by ☺.

3.1 Foundations

Review of decision theory

- ☺Robert, C. (2007). *The Bayesian choice: from decision-theoretic foundations to computational implementation*. Springer Verlag, chapter 2.

Machine learning using Python

- VanderPlas, J. (2022). *Python Data Science Handbook*. O'Reilly, chapter 5.

Probably approximately correct learning theory

- ☺Shalev-Shwartz, S. and Ben-David, S. (2014). *Understanding machine learning: From theory to algorithms*. Cambridge University Press, chapters 2 to 6.

Shrinkage in the normal means model

- Wasserman, L. (2006). *All of nonparametric statistics*. Springer Science & Business Media, chapter 7.

- ☺Stigler, S. M. (1990). The 1988 Neyman memorial lecture: a Galtonian perspective on shrinkage estimators. *Statistical Science*, pages 147–155.

- ☺Morris, C. N. (1983). Parametric empirical Bayes inference: Theory and applications. *Journal of the American Statistical Association*, 78(381):pp. 47–55.

- Stein, C. M. (1981). Estimation of the mean of a multivariate normal distribution. *The Annals of Statistics*, 9(6):1135–1151.

- van der Vaart, A. W. (2000). *Asymptotic statistics*. Cambridge University Press, chapter 7.

- Hansen, B. E. (2016). Efficient shrinkage in parametric models. *Journal of Econometrics*, 190(1):115–132.

- Abadie, A. and Kasy, M. (2019). Choosing among regularized estimators in empirical economics - the risk of machine learning. *Review of Economics and Statistics*, 101(5).

- Fessler, P. and Kasy, M. (2019). How to Use Economic Theory to Improve Estimators: Shrinking Toward Theoretical Restrictions. *The Review of Economics and Statistics*, 101(4):681–698.

3.2 Supervised learning

Gaussian process priors, reproducing kernel Hilbert spaces, and Splines

- ☺Williams, C. and Rasmussen, C. (2006). *Gaussian processes for machine learning*. MIT Press, chapters 2 and 7.
- Wahba, G. (1990). *Spline models for observational data*, volume 59. Society for Industrial Mathematics, chapter 1.
- Kasy, M. (2016). Why experimenters might not always want to randomize, and what they could do instead. *Political Analysis*, 24(3):324–338.
- Kasy, M. (2018). Optimal taxation and insurance using machine learning – sufficient statistics and beyond. *Journal of Public Economics*, 167.

Regression trees and random forests

- ☺Friedman, J., Hastie, T., and Tibshirani, R. (2001). *The elements of statistical learning*, volume 1. Springer series in statistics Springer, Berlin, chapters 8 and 9.
- Athey, S. and Imbens, G. (2016). Recursive partitioning for heterogeneous causal effects. *Proceedings of the National Academy of Sciences*, 113(27):7353–7360.

Deep neural nets

- ☺Goodfellow, I., Bengio, Y., and Courville, A. (2016). *Deep learning*. MIT Press, chapters 6-8.
- Bottou, L., Curtis, F. E., and Nocedal, J. (2018). Optimization methods for large-scale machine learning. *SIAM Review*, 60(2):223–311
- Bartlett, P. L., Montanari, A., and Rakhlin, A. (2021). Deep learning: a statistical viewpoint. *arXiv preprint arXiv:2103.09177*

Large Language Models and Transformers

- Jurafsky, D. and Martin, J. H. (2023). *Speech and Language Processing*. <https://web.stanford.edu/~jurafsky/slp3/>, chapters 9–11.

Image generation, autoencoders, and diffusion models

- Luo, C. (2022). Understanding diffusion models: A unified perspective. *arXiv preprint arXiv:2208.11970*
- Yang, L., Zhang, Z., Song, Y., Hong, S., Xu, R., Zhao, Y., Zhang, W., Cui, B., and Yang, M.-H. (2023). Diffusion models: A comprehensive survey of methods and applications. *ACM Computing Surveys*, 56(4):1–39

Double/debiased machine learning

Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., and Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68.

Conformal inference

Angelopoulos, A. N. and Bates, S. (2021). A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *arXiv preprint arXiv:2107.07511*

3.3 Online learning and active learning

Online learning

☺Cesa-Bianchi, N. and Lugosi, G. (2006). *Prediction, learning, and games*. Cambridge University Press, chapter 2.

Lewi, Y., Kaplan, H., and Mansour, Y. (2020). Thompson sampling for adversarial bit prediction. In *Algorithmic Learning Theory*, pages 518–553. PMLR

Hazan, E. (2016). Introduction to online convex optimization. *Foundations and Trends in Optimization*, 2(3-4):157–325.

Bandit problems

Bubeck, S. and Cesa-Bianchi, N. (2012). Regret Analysis of Stochastic and Nonstochastic Multi-armed Bandit Problems. *Foundations and Trends® in Machine Learning*, 5(1):1–122.

☺Russo, D. J., Roy, B. V., Kazerouni, A., Osband, I., and Wen, Z. (2018). A Tutorial on Thompson Sampling. *Foundations and Trends® in Machine Learning*, 11(1):1–96.

Weber, R. et al. (1992). On the Gittins index for multiarmed bandits. *The Annals of Applied Probability*, 2(4):1024–1033.

☺Kasy, M. and Sautmann, A. (2021). Adaptive treatment assignment in experiments for policy choice. *Econometrica*, 89(1):113–132.

Wager, S. and Xu, K. (2021). Diffusion asymptotics for sequential experiments. *arXiv preprint arXiv:2101.09855*.

Caria, S., Gordon, G., Kasy, M., Osman, S., Quinn, S., and Teytelboym, A. (2024). Job search assistance for refugees in Jordan: An adaptive field experiment. *Journal of the European Economic Association*, 22(2).

Cesa-Bianchi, N., Colomboni, R., and Kasy, M. (2025). Adaptive maximization of social welfare. *Econometrica*, 93(3).

Reinforcement learning

Sutton, R. S. and Barto, A. G. (2018). *Reinforcement learning: An introduction*. MIT press.

François-Lavet, V., Henderson, P., Islam, R., Bellemare, M. G., and Pineau, J. (2018). An introduction to deep reinforcement learning. *Foundations and Trends® in Machine Learning*, 11(3-4):219–354.

3.4 Ethics and machine learning

☺Kearns, M. and Roth, A. (2019). *The Ethical Algorithm: The Science of Socially Aware Algorithm Design*. Oxford University Press.

☺Kasy, M. (2025). *The means of prediction: How AI Really Works (and Who Benefits)*. University Of Chicago Press.

Fairness

Pessach, D. and Shmueli, E. (2020). Algorithmic fairness. *arXiv preprint arXiv:2001.09784*

Kasy, M. and Abebe, R. (2021). Fairness, equality, and power in algorithmic decision making. *ACM Conference on Fairness, Accountability, and Transparency*.

Kasy, M. (2024). Algorithmic bias and racial inequality: A critical review. *Oxford Review of Economic Policy*.

Roemer, J. E. (1998). *Theories of distributive justice*. Harvard University Press, Cambridge.

Differential privacy

Dwork, C. and Roth, A. (2014). The algorithmic foundations of differential privacy. *Foundations and Trends® in Theoretical Computer Science*, 9(3-4):211–407.

Explainability

Lipton, Z. C. (2018). The mythos of model interpretability: In machine learning, the concept of interpretability is both important and slippery. *Queue*, 16(3):31–57.

Wachter, S., Mittelstadt, B., and Russell, C. (2018). Counterfactual explanations without opening the black box: automated decisions and the GDPR. *Harvard Journal of Law and Technology*, 31(2).